Knowledge Based Engineering (KBE) AE4204

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Lecture 2

KBE and the Artificial Intelligence roots

- Frame Based Systems (part 2)

- Object Oriented paradigm

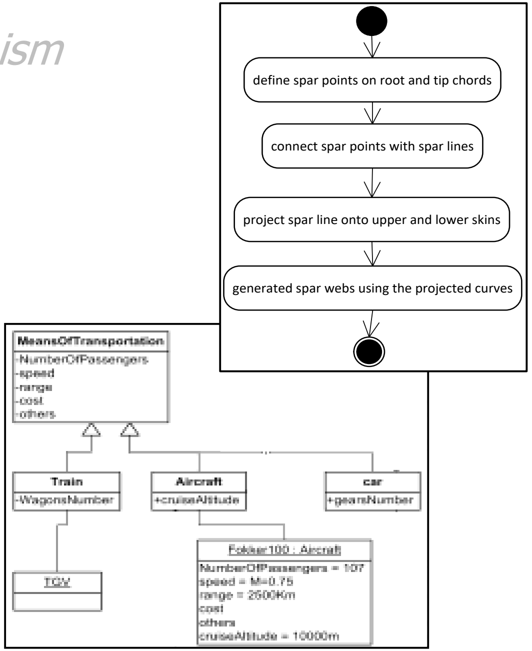
- UML

Contents

- A definition of Knowledge Based Engineering (KBE)
- Knowledge Based Systems (KBS)
  - Working principles: *production rules* and *inference mechanism*
- Frame Based Systems (FBS)
  - The Object-Oriented paradigm applied to KBS

+ Object-Oriented modeling concepts

+ Unified Modeling Language (UML)



Frame Based Systems

The object-oriented paradigm applied to knowledge based systems.

Class-frame

Aircraft	
MTOW	Kg
No of passengers	
Wing span	m
Total length	m
Wing area	m^2
Range	Km

Fokker100: Aircraft	
MTOW	43000 Kg
No of passengers	107
Wing span	28 m
Total length	35.5 m
Wing area	93.5 m^2
Range	2500 Km

possible **instance-frames** of the same class-frame

AirbusA380: Aircraft	
MTOW	560000 Kg
No of passengers	853
Wing span	79,8 m
Total length	73 m
Wing area	845 m^2
Range	15200 Km

Frame Based Systems

The object-oriented paradigm applied to knowledge based systems.

Class-frames to define **generic types of objects**

E.g. Class-frame *Aircraft*

Instance (or object)-frames to define **specific instances** of a generic class

E.g. The object-frame *fokker100* is a type of *Aircraft*

How to instantiate a class-frame?

By specifying a name to the object-frame and assigning **values** to the class-frame slots.

→ Any given set of slot values yields a <u>different</u> and <u>specific</u> instance of the <u>same</u> class-frame

Frame Based Systems

The object-oriented paradigm applied to knowledge based systems.

The concepts of **class**-frames and **instance**-frames are not different from the concepts of **classes** and **objects** in the **Object Oriented (OO) paradigm***.

According to the OO modelling approach, each entity (or object) is a **unique** instantiation of a generic class.

A class is a model for a **type of** objects. It is the **blueprint**, or **template**, or **factory** from which individual objects are created

Frame Based Systems

The object-oriented paradigm applied to knowledge based systems.

It is possible to define a class as a **specialization** of a more generic class → **superclass**.

E.g. the class *Aircraft* can be defined as a specialization of the superclass *MeansOfTransportation*

When a class is defined as a specialization of a superclass, it automatically **inherits all the attributes** of the superclass*.

Next to the inherited attributes, a (sub)class generally features **extra attributes** (which are making it a different specialization of the superclass).

The (sub)class might also **override** some superclass attributes.

Frame Based Systems

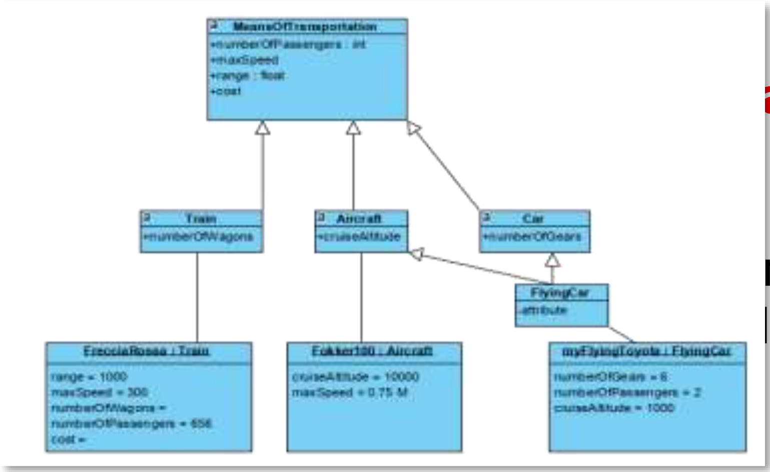
The object-oriented paradigm applied to knowledge based systems.

By means of the concepts of **object, class**, **superclass** and relationships such as **is-a-type-of** and **is-a-specialization-of**, it is already possible to model complex bodies of knowledge.

How to present these models in a rigorous but visual way to improve communication and knowledge sharing?

→ **The Unified Modeling Language (UML)**

The Unified Modeling Language (UML)



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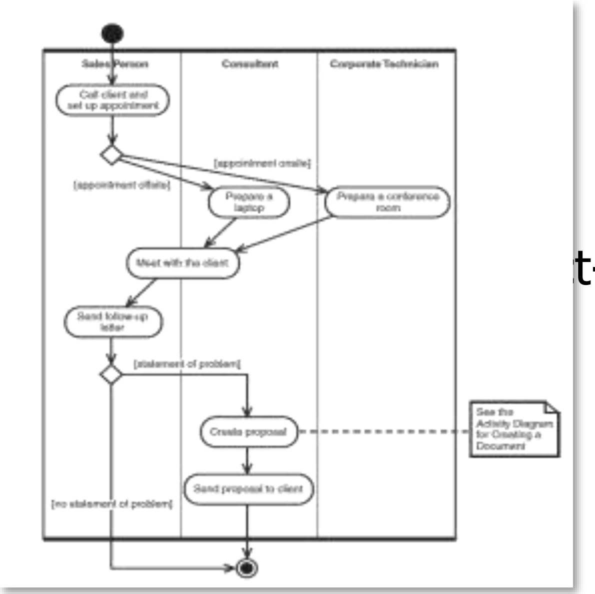


UML is a toolbox of diagram types, generally di

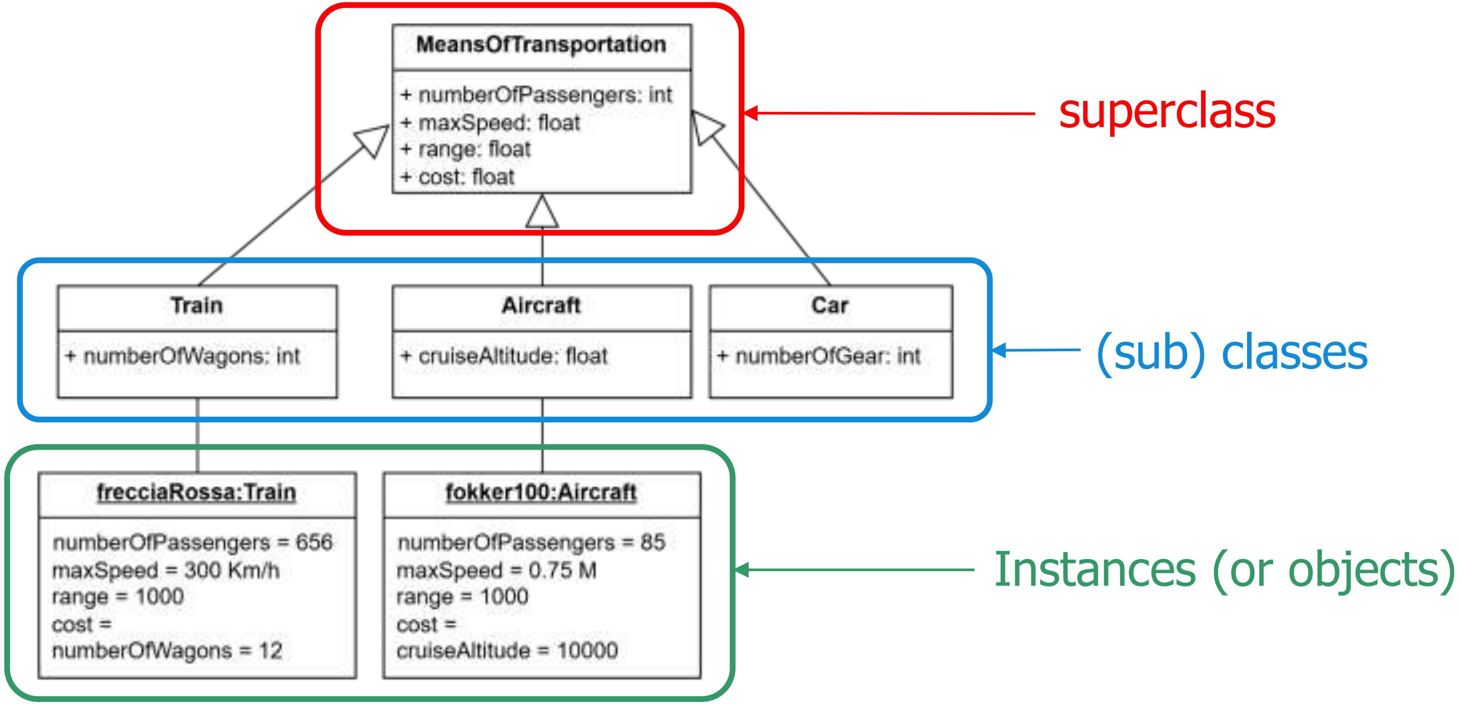
1. Structural diagrams, e.g. **class diagrams**, architecture of a system
2. Behavioral diagram, e.g. **activity diagrams**, to describe how the system behaves

Unified Modeling Language (UML)

- Class diagrams
- Concepts:
  - Classes & Superclasses
  - Objects
  - Inheritance

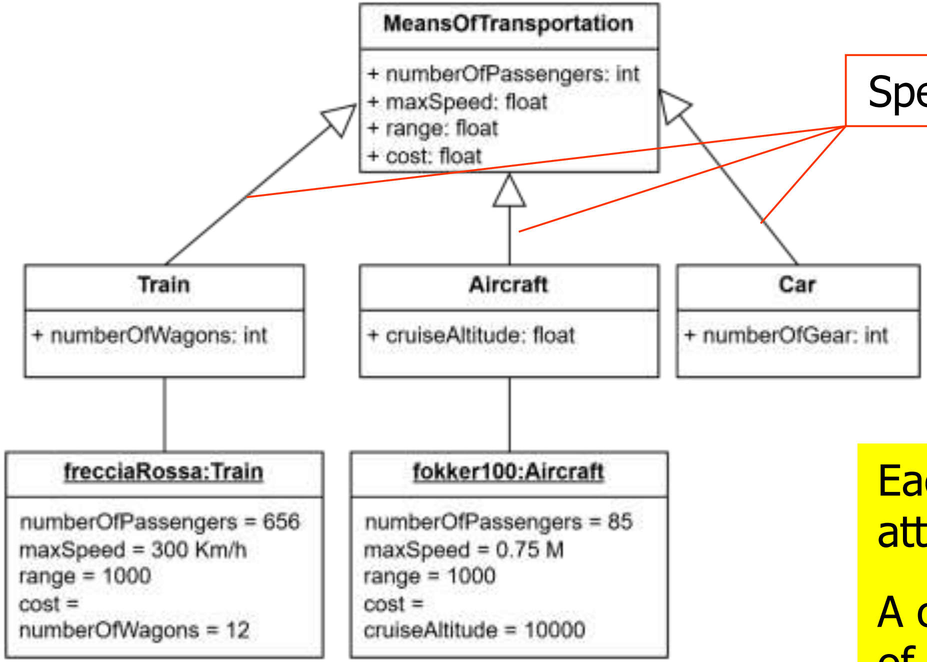
UML diagrams to illustrate OO modeling

The class diagram and the specialization/inheritance link



UML diagrams to illustrate OO modeling

The class diagram and the specialization/inheritance link



Specialization/***inheritance*** link (*is a*)

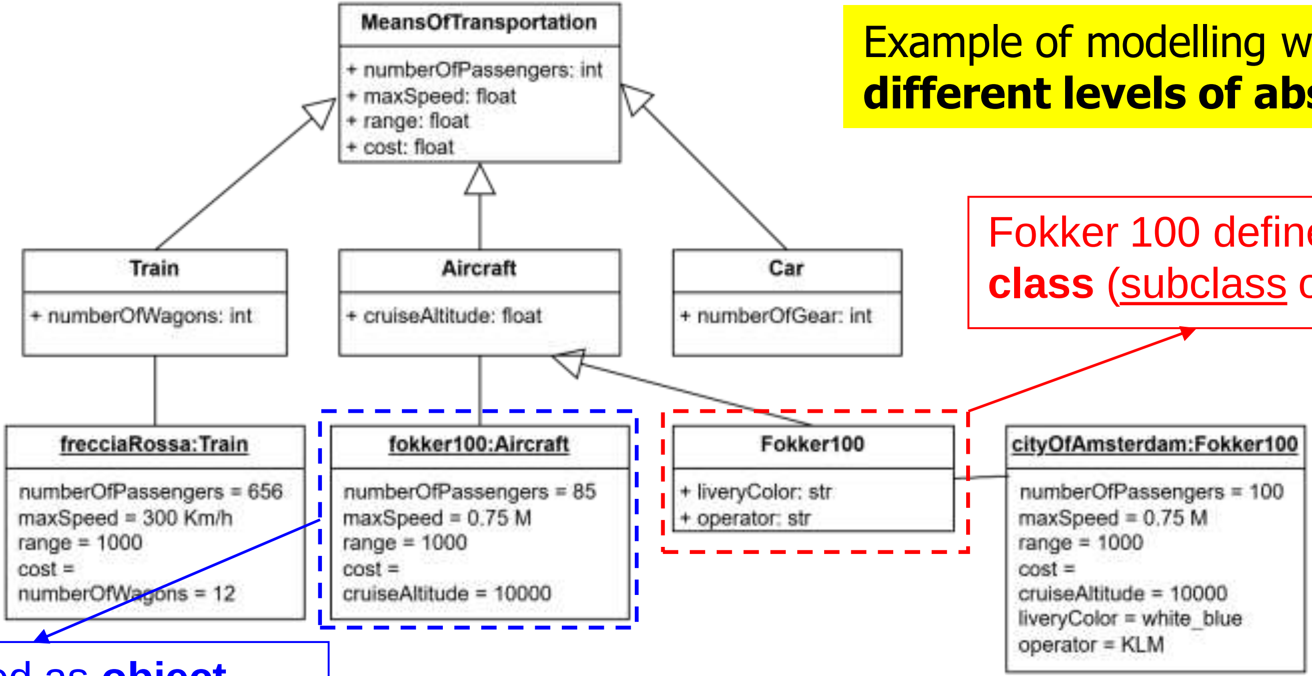
Each class ***inherits*** the list of attributes from its superclass.

A class adds and/or overwrites some of the attributes of its superclass

UML diagrams to illustrate OO modeling

The class diagram and the specialization/inheritance link

Example of modelling with
different levels of abstraction



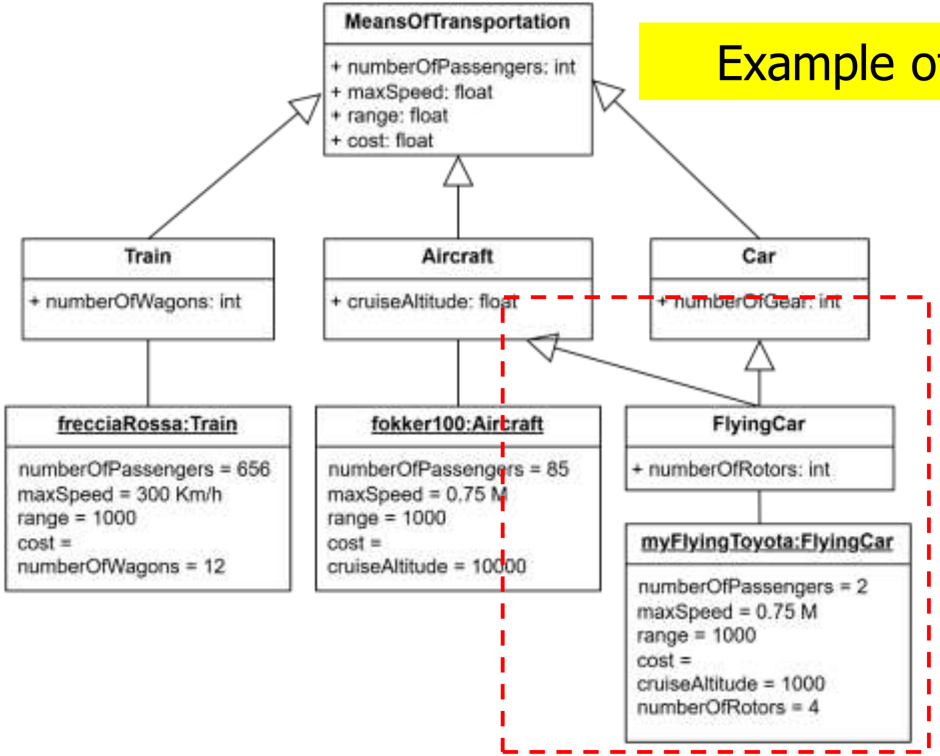
Fokker 100 defined as
class (<u>subclass</u> of Aircraft)

Fokker 100 defined as **object**
(<u>instantiation</u> of the class Aircraft)

UML diagrams to illustrate OO modeling

The class diagram and the specialization/inheritance link

Example of **multiple inheritance**



UML diagrams to illustrate OO modeling

The class diagram and the specialization/inheritance link

Try the following:

1. Define the class Smartphone
2. Define a superclass for the Smartphone class
3. Define Smartphone as superclass and define a class that inherits from it
4. Define an instance of Smartphone

Are you speaking UML? Check the following:

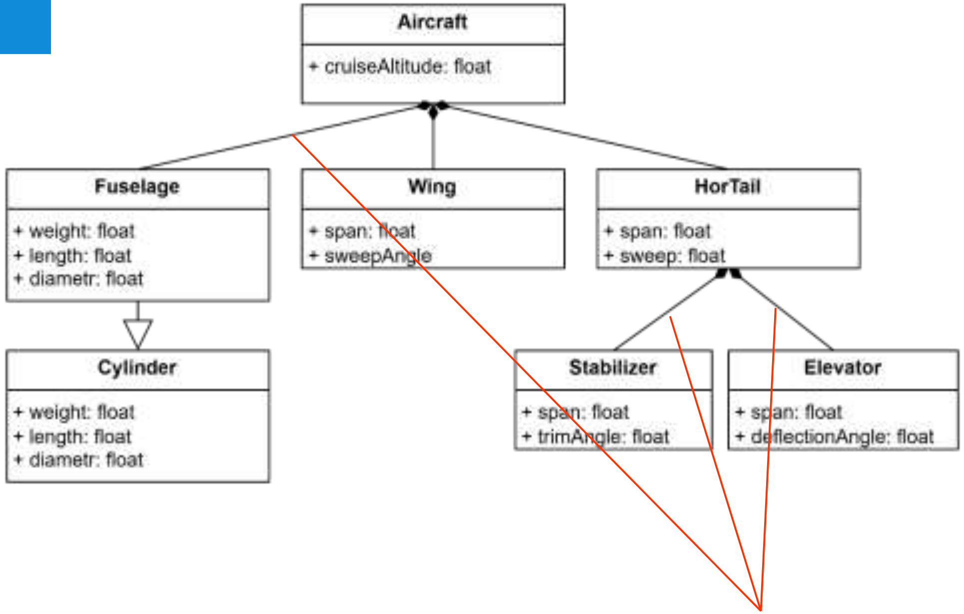
- *Are you using the right graphical syntax to model a class/superclass/instance?*
- *Are you using the right graphical syntax to model relationships between classes/superclasses/instances?*

The AI origin of KBE

- **Frame based systems (continued)**

Frame Based Systems.

The object oriented paradigm applied to knowledge based systems.



The OO modeling approach allows also the representation of ***part-whole*** **systems**, i.e. hierarchies of concepts based on the relationship ***has-part***

A frame slot can contain **pointers** to other frames, or **to specific slots** of other frames

UML representation of ***Composition links*** (i.e. *has-part*)

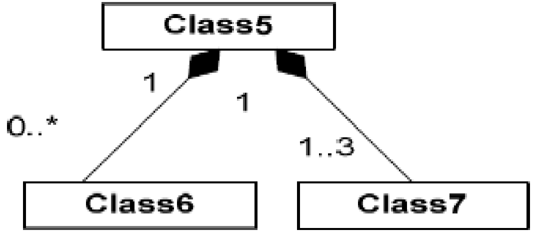
Unified Modeling Language (UML)

- Class diagrams
  - Aggregations/compositions

UML diagrams to illustrate OO modeling

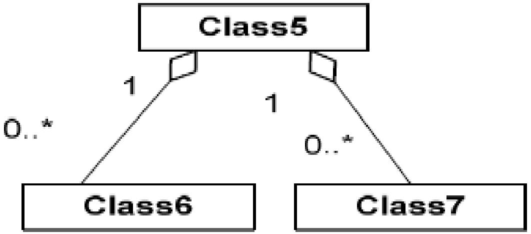
UNIFIED MODELING LANGUAGE

The class diagram and the composition and aggregation links



Composition link: Class6 and Class7 are components of Class5.

Multiplicity label: Class 5 is a composition including from 0 to an infinite number [0..*] of Class6 components; and 1 to 3 [1..3] Class7 components. Both Class6 and Class7 belong to only 1 Class5 composition [1].

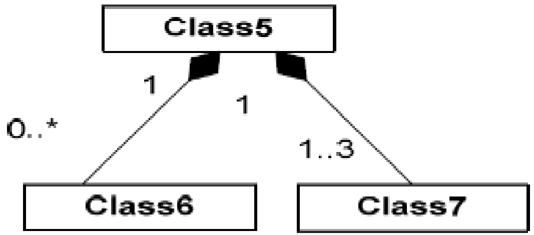


Aggregation link: Class6 and Class7 are members of the aggregation Class5.

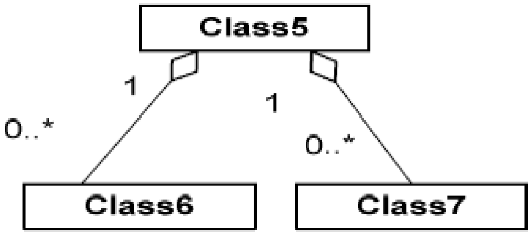
*A **composition** is a stronger relation than **aggregation**: all the members of an aggregation can live autonomously without the need of the aggregation class. On the contrary, if a composition is eliminated, all the members are also eliminated.*

UML diagrams to illustrate OO modeling
The class diagram and the composition and aggregation links

composition



aggregation



Search for **aggregation** and **composition** relationships!

UML diagrams to illustrate OO modeling

The class diagram and the composition and aggregation links

Try the following:

1. Add some component classes to your Smartphone class
2. Indicate some multiplicity labels
3. Think of possible example of composition and aggregations

To summarize...

- Frame Based Systems offer a more advanced domain knowledge representation than rule-based systems, featuring concepts from the **Objected Oriented modeling paradigm**

- The concepts of **class-frame** allows defining *few* conceptual templates which can be reused (*instantiated*) to model *many* individual **instance-frames**

- Frames can be connected* to other frames by means of **inheritance** and **composition/aggregation** links, thereby allowing complex-yet-efficient domain knowledge representations

- The UML (**Unified Modeling Language**), an industry-standard visual language specifically developed to support OO modeling, offers very effective diagrammatic representation of systems, their components and relationships

…and to anticipate

- Frame slots can contain **pointers to slots of other frames**
- Frame slots can contain **procedures (*methods*)** to dynamically compute their values
- These procedures may include **IF-THEN** statements, as seen in ruled based KBSs*
- The evaluation of these procedures can be **eager** or **on-demand (lazy)**
- Blurred separation between knowledge base and inference mechanism

Main reference material for this lecture

- **Chapter 3** *Knowledge Based Engineering. The AI roots and the OO paradigm*

- **Journal paper** *(G. La Rocca, Knowledge based engineering: Between AI and CAD. Review of a language based technology to support engineering design, Advanced Engineering Informatics, 26 (2012) pp159–179*

- **On UML** (*Object Oriented modeling Class and Object diagrams)*

- **On UML** (*A summary of basic UML notations)*

All available in the section *Reading Material* on